

Historical Incident: Carbonate Losses & Differential Sticking in MH-112 (illustrative)

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1. Incident Overview

During exploration drilling on Mumbai High reference well MH-112 in fractured Miocene limestone, total fluid losses of 45 bbl/hr occurred when drilling overbalanced through a depleted vuggy zone at 2,850 m MD. Continued pipe manipulation while attempting to mix LCM led to differential sticking of the 8-1/2 in BHA against permeable limestone filter cake.

2. Sequence of Events

- ****Day 14 04:30****: Drilling in limestone at 2,850 m MD with 10.80 ppg mud. Sudden loss of returns observed; pit level dropped 35 bbl in 45 minutes.
- ****Day 14 05:20****: Stopped pumps to perform dynamic loss assessment. Static losses stabilized at 20 bbl/hr.
- ****Day 14 06:15****: Attempted to pull off bottom; string became differentially stuck with 80 klbf overpull.
- ****Day 14 11:00****: Spotted 50 bbl hydrocarbon-based freeing pipe pill around BHA.
- ****Day 15 02:00****: String jarred free after 14.5 hours of jarring operations. Total NPT incurred: 48.0 hours.

3. Lessons for Deepwater Project Samudra Manthan

- In transition zones and depleted sections, overbalance must be tightly constrained to prevent differential sticking.
- When drilling narrow margins in MN-SM-DW-01, maintaining ECD strictly within the 11.48 ppg pore pressure to 12.10 ppg shoe FIT envelope is essential to prevent both kicks and severe loss-induced pipe sticking.